MODULE-V
ADVANCES IN MECHANICAL ENGINEERING

Automation technology: Definition of automation, types of automation

Basic elements of automation.

Automation Technology

- Automation can be defined as the technology by which a process or procedure is accomplished without human assistance.
- It is implemented using a program of instructions combined with a control system that executes the instructions.
- To automate a process, power is required, both to drive the process itself and to operate the program and control system.

Types of automation

1) Fixed Automation:
   Automation used for high-volume production with a fixed sequence of operations.
2) Programmable Automation:
   Automation in which the production process is controlled by a program that can be changed for different products.
3) Flexible Automation:
   Automation system capable of producing different products with minimal or no setup time.
4) Integrated Automation:
   Automation where design, manufacturing, and control systems are fully computer integrated.
5) Robotic Automation:
   Use of industrial robots to perform repetitive and precise manufacturing tasks.

Basic elements of automation

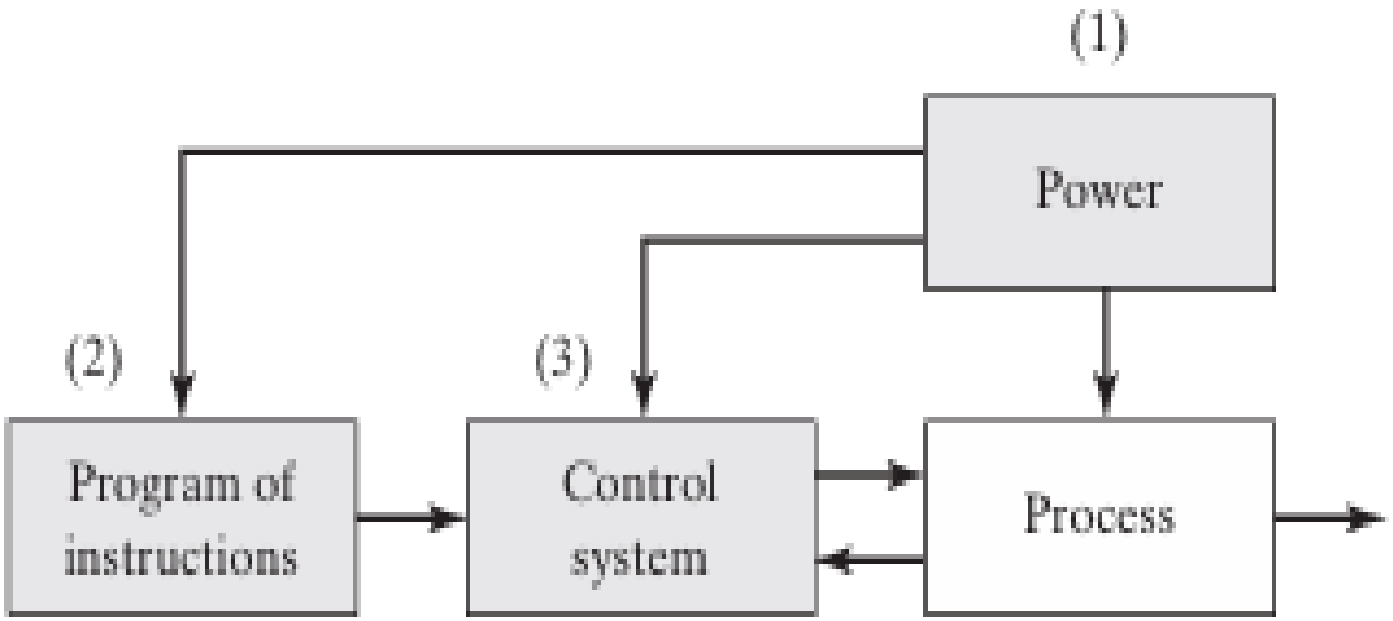


Figure 4.2 Elements of an automated system: (1) power, (2) program of instructions, and (3) control systems.

Basic elements of automation

1) Power:
   Supplies the energy required to operate the process and control system.
2) Program of Instructions:
   A set of coded commands that defines the sequence of operations to be performed by the system.
3) Control System:
   Interprets the program instructions and controls the process by sending signals to actuators and receiving feedback.
4) Process:
   The actual operation or manufacturing activity carried out to produce the desired output.

Mechatronics systems: Definition of mechatronics, elements of mechatronics systems, Examples.

Definition of mechatronics

Mechatronics can be defined as an interdisciplinary approach to engineering design through an integration of Mechanical engineering, Electronics engineering and Electrical engineering.

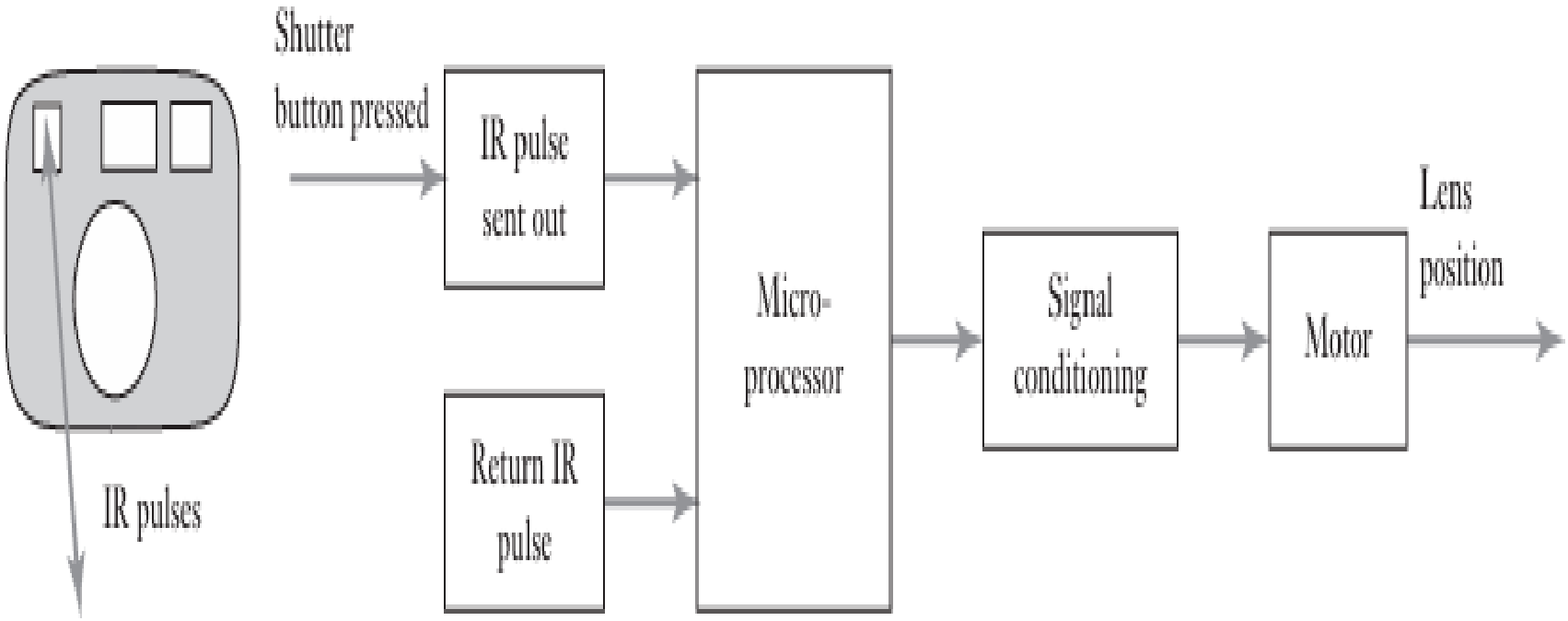
Mechatronics brings together areas of technology involving

- Sensor and measurement systems
- Control systems
- Microprocessor system
- Drive and actuation systems

Elements of mechatronics systems

1) Mechanical System:
   Physical components such as gears, shafts, springs, and structures.
2) Sensors & Transducers:
   Convert physical quantities (position, pressure, temperature) into electrical signals.
3) Actuators:
   Convert electrical signals into mechanical motion (motors, solenoids, cylinders).
4) Control System / Controller:
   Processes sensor signals and generates control actions (microcontroller, PLC).
5) Signal Conditioning:
   Modifies sensor signals to suitable levels (amplifiers, filters).
6) Power Supply:
   Provides required electrical, hydraulic, or pneumatic power.
7) Software:
   Programs that control system logic, sequencing, and decision-making.

Examples of mechatronics systems (Digital camera and autofocus)



Examples of mechatronics systems (Digital camera and autofocus)

- A digital camera is likely to have an autofocus control system.
- When the photographer presses the shutter button, a transducer on the front of the camera sends pulses of infrared (IR) light towards the subject of the photograph.
- The infrared pulses bounce off the subject and are reflected back to the camera where the same transducer picks them up.
- For each meter the subject is distant from the camera, the round-trip is about 6 m. The time difference between the output and return pulses is detected and fed to a microprocessor. This has a set of values stored in its memory and so gives an output which rotates the lens housing and moves the lens to a position where the object is in focus.
- This type of autofocus can only be used for distances up to about 10 m as the returning infrared pulses are too weak at greater distances.

Elementary sensors: Working principle
and Applications of Potentiometer Optical encoders.

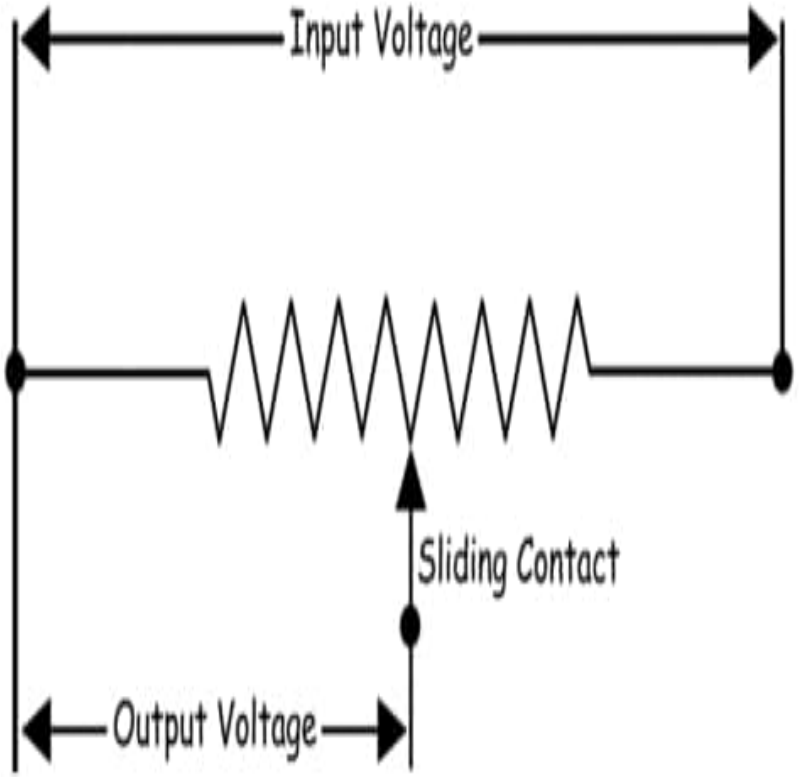
Elementary sensors

- The term sensor is used for an element which produces a signal relating to the quantity being measured.
- An electrical resistance temperature element, the quantity being measured is temperature and the sensor transforms an input of temperature into a change in resistance.

Potentiometer

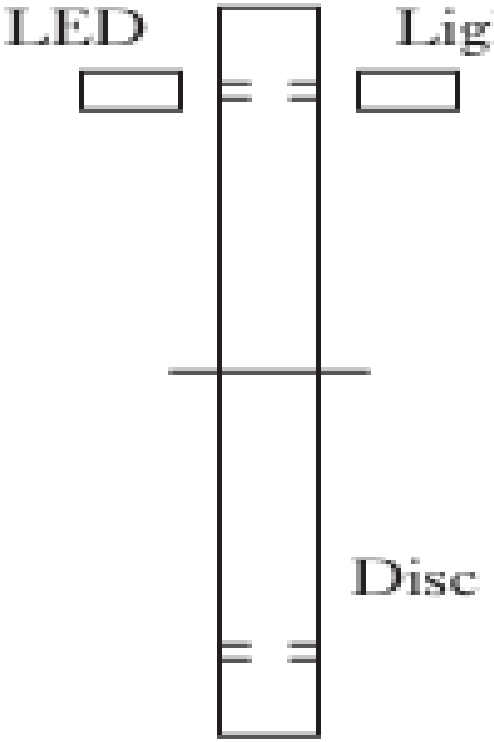
- A potentiometer is defined as a 3 terminal variable resistor in which the resistance is manually varied to control the flow of electric current. A potentiometer acts as an adjustable voltage divider.
- Types of Potentiometers: There are two main types of potentiometers: rotary, which adjusts by rotating, and linear, which adjusts by sliding linearly.
- Applications: Potentiometers are used for comparing the emf of battery cells, measuring internal resistance, and adjusting voltages in various circuits.

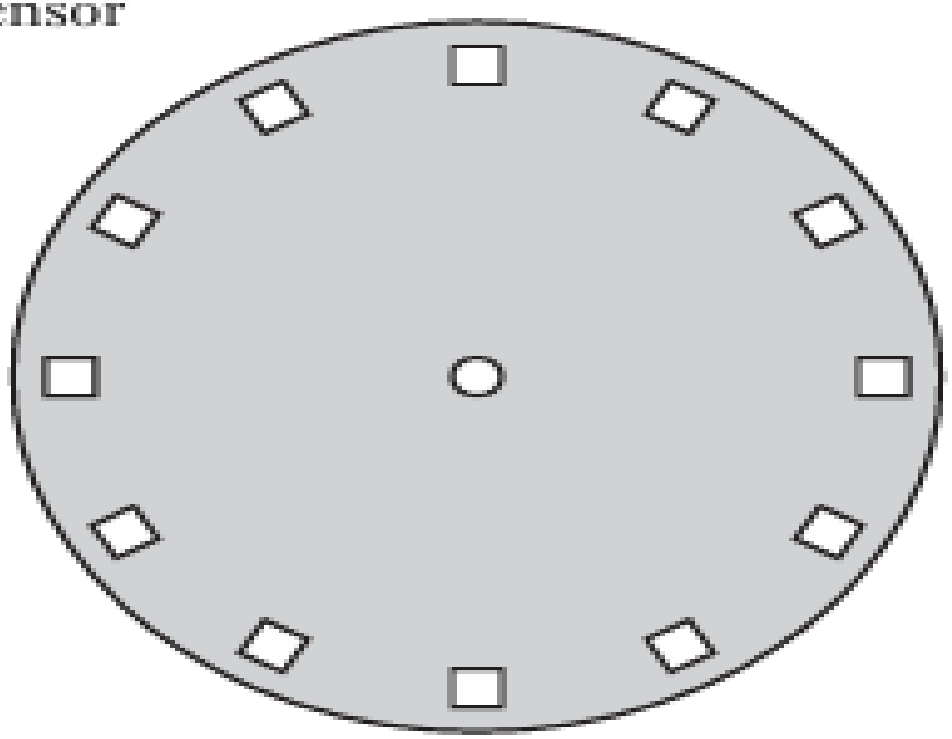
Working Principle of Potentiometers



- Potentiometers work by varying the position of a sliding contact across a uniform resistance.
- In a potentiometer, the entire input voltage is applied across the whole length of the resistor, and the output voltage is the voltage drop between the fixed and sliding contact.
- A potentiometer has the two terminals of the input source fixed to the end of the resistor. To adjust the output voltage the sliding contact gets moved along the resistor on the output side.

Optical encoder





The disc with slots through which the light can pass

Optical encoder

- An encoder is a device that provides a digital output as a result of a linear or angular displacement. Position encoders can be grouped into two categories: incremental encoders, which detect changes in rotation from some datum position; and absolute encoders, which give the actual angular position.
- The basic form of an incremental encoder for the measurement of angular displacement. A beam of light passes through slots in a disc and is detected by a suitable light sensor. When the disc is rotated, a pulsed output is produced by the sensor with the number of pulses being proportional to the angle through which the disc rotates. Thus the angular position of the disc, and hence the shaft rotating it, can be determined by the number of pulses produced since some datum position.

Integrated system: Need for integration of technologies, ADAS (Advanced Driver Assistance System).

Need for Integration of Technologies

- The integration of technologies means combining different technologies (mechanical, electrical, electronics, computer, and control systems) to work together as a single system.
- It is needed for the following reasons.

1) Improved system performance – increases accuracy, speed, and efficiency
2) Automation of processes – reduces manual work and human error
3) Compact and efficient designs – combines multiple functions into one system
4) Better control and monitoring – real-time data acquisition and feedback
5) Higher productivity – faster operations with consistent quality
6) Flexibility and adaptability – easy modification and upgrading of systems
7) Cost reduction – lowers production, maintenance, and operating costs
8) Reliability and safety – integrated systems provide fault detection and protection

ADAS
(Advanced Driver Assistance System).

- Safety systems which are designed to assist the driver in both normal driving modes and pre-collision avoidance. These systems rely on the use of numerous specialized sensors and cameras that often require calibration in order to operate properly. Calibration can be static, dynamic or a combination of the two.
- Advanced driver assistance system functions are categorized into:
- Passive functions – the vehicle will warn the driver in the form of a warning light, warning message, an audible indication (tone) or feedback (vibration) when sensing potentially dangerous situations.
- Active functions – the vehicle can take corrective actions when sensing potentially dangerous situations.